

Problem Solving Using Python (CSE1012)

Dr. Abdul Kayom, Assistant Professor, SENSE

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Documentation String

- It serves the similar purpose as that of comments
- It is designed to explain code
- However, it is more specific and has proper syntax

Documentation String

#Example of Document Strings inside a Function

```
def func():
```

```
    """The Program just prints a message.
```

```
        It will display: Hello World!! """
```

```
    print("Hello World!!")
```

```
help(func)
```

```
func()
```

Documentation String

Help on function func in module `__main__`:

`func()`

The Program just prints a message.

It will display: Hello World!!

Hello World!!

Example Program

- Compute the Series: $1/1! + 4/2! + 27/3! + \dots$

Example Program

#Calculate the series: $1/1! + 4/2! + 27/3!$

```
def fact(n):  
    f = 1  
    if(n==0 or n==1):  
        return 1  
    else:  
        for i in range(1,n+1):  
            f=f*i  
        return f  
  
n = int(input("Enter the Number: "))  
s = 0  
for i in range(1,n+1):  
    s = s + (i**i/fact(i))  
print("The Sum of the Series is: ",s)
```

Example Program

- Calculate the total number of Matches to be played if the number of teams = 10, each team play against each other only once.

Example Program

- Calculate the total number of Matches to be played if the number of teams=10, each team play against others only once.
- $nCr = n! / (r! * (n-r)!)$

Example Program

- Calculate the total number of Matches to be played if the number of teams=10, each team play against others only once.
- $nCr = n! / (r! * (n-r)!)$
- Simplified formula is: $nC2 = n(n-1)/2$

Example Program

#Number of Matches played if each team play each other only once

```
def fact(n):
```

```
    f=1
```

```
    if(n==0 or n==1):
```

```
        return 1
```

```
    else:
```

```
        for i in range(1,n+1):
```

```
            f=f*i
```

```
        return f
```

```
n=int(input("Enter the Number of teams: "))
```

```
tot=int(fact(n)/(fact(2)*fact(n-2)))
```

```
print("The total number of matches to be played are: ",tot)
```

Example Program

- Compute nPr
- $nPr = n!/(n-r)!$

Example Program

#Compute Permutation nPr

```
def fact(n):
```

```
    f=1
```

```
    if(n==0 or n==1):
```

```
        return 1
```

```
    else:
```

```
        for i in range(1,n+1):
```

```
            f=f*i
```

```
        return f
```

```
n=int(input("Enter the value of n: "))
```

```
r=int(input("Enter the value of r: "))
```

```
nPr=int(fact(n)/(fact(n-r)))
```

```
print("nPr is: ",nPr)
```

Example Program

- Compute nCr
- $nCr = n! / (r! * (n-r)!)$

Example Program

#Compute Combination nCr

```
def fact(n):  
    f=1  
    if(n==0 or n==1):  
        return 1  
    else:  
        for i in range(1,n+1):  
            f=f*i  
        return f  
n=int(input("Enter the value of n: "))  
r=int(input("Enter the value of r: "))  
nCr=int(fact(n)/(fact(r)*fact(n-r)))  
print("nCr is: ",nCr)
```

Thank You

Dr. Abdul Kayom



Asst. Prof, (SENSE)



kayomabdul@vitap.ac.in



8474860025